

Geography Ch-8 (Advanced) — Limestone Caves of India & the Meghalayan Age

Advanced / India-applied | GS-I & GS-III | Source: D.R. Khullar (India: A Comprehensive Geography)

India's karst caves, cave tourism, stalactite/stalagmite formations

Extra CA: Mawmluh Cave / Meghalayan Age GSSP (distinct from Must-Do's mining-aquifer CA)

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

ADVANCED — India's Limestone Caves & Cave Tourism (Khullar)

GS-I · India
Geography →
Karst / Caves
(Source: D.R.
Khullar)

■ NEWS TRIGGER

India's finest karst caves lie in Meghalaya's Jaintia and Garo hills — including Asia's longest cave (~19.2 km) and 200+ others — a major boost to geological cave tourism.

■ INTRO & ORIGIN

Khullar highlights India's limestone caves both as geomorphic showcases (stalactite/stalagmite) and as cultural/tourism assets, from Meghalaya to Ajanta and Pachmarhi.

The Meghalaya caves form in thick limestone under very high rainfall (Cherrapunji) — ideal conditions for rapid solution and dripstone growth.

■ KEY DATA

Site	State	Note (Khullar)
Jaintia Hills cave	Meghalaya	Asia's longest cave ~19.2 km; 200+ caves
Cherrapunji (Garo/Khasi) cave	Meghalaya	Limestone cave; well-developed stalactite/stalagmite
Ajanta (of ~30 caves)	Maharashtra	Most popular of the Aurangabad caves
Pachmarhi (~500 caves)	Madhya Pradesh	Some with early-man paintings
Khandagiri & Udayagiri	Odisha	Twin hills; rock sculpture caves

■ STATIC THEORY

- Meghalaya has India's premier **limestone (karst) cave systems** — the Jaintia hills hold Asia's longest cave (~19.2 km) plus 200+ others.
- The **Cherrapunji** limestone cave shows well-developed **stalactite and stalagmite** formations (dripstone deposition).
- Cave tourism spans geology + culture: **Ajanta** (Aurangabad), **Pachmarhi** (~500 caves, early-man paintings), **Khandagiri/Udayagiri** (Odisha).
- Very high rainfall over thick limestone makes Meghalaya ideal for rapid **solution** and cave growth.
- These caves double as **palaeoclimate archives** — their stalagmites record past rainfall/drought.

■ MUST-KNOW FACTS

- Asia's longest cave (~19.2 km) is in the Jaintia hills, Meghalaya.
- Cherrapunji limestone cave: developed stalactite/stalagmite (Khullar).
- Ajanta (Maharashtra), Pachmarhi (MP ~500 caves), Khandagiri/Udayagiri (Odisha).
- High rainfall + thick limestone => rapid karst solution in Meghalaya.

■ UPSC TRAPS

- ✗ India has no significant natural limestone caves.
- ✓ Meghalaya hosts India's largest natural karst cave systems, including Asia's longest cave.
- ✗ Ajanta and Pachmarhi are natural solution caves.
- ✓ Many are rock-cut/cultural caves; the classic natural karst caves are in Meghalaya.

■ MAINS ANGLE

■ STUDY LINK

Assess Meghalaya's karst caves as geomorphic, palaeoclimatic and tourism assets, and the need to protect them from mining and unregulated tourism.

Geography → India → Karst Caves & Cave Tourism (Khullar: Tourism / Physiography)

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA ANCHOR — Mawmluh Cave & the Meghalayan Age (GSSP)

GS-I • Karst
applied →
Geological time
scale

■ NEWS TRIGGER

A stalagmite (KNI-51) in Mawmluh Cave near Cherrapunji is the global reference point (GSSP) defining the 'Meghalayan Age' — the current stage of the Holocene, begun ~4,200 years ago.

■ INTRO & ORIGIN

Karst dripstone is a precise climate recorder: Mawmluh's stalagmite captured a global ~4,200-year-ago mega-drought, earning it a place on the geological time scale.

■ STATIC THEORY

- **GSSP** = Global Boundary Stratotype Section and Point — the physical 'golden spike' defining a time boundary.
- The **Meghalayan Age** is the youngest stage of the Holocene (last ~4,200 years); its GSSP is the KNI-51 stalagmite in **Mawmluh Cave**, Meghalaya.
- The stalagmite's chemistry records a global drought (~4,200 BP) linked to the collapse of several early civilisations.

■ MUST-KNOW FACTS

1. Meghalayan Age = current stage of the Holocene (~last 4,200 yrs).
2. GSSP defined by stalagmite KNI-51 in Mawmluh Cave, Meghalaya.
3. Basis: a global ~4,200-year-ago mega-drought recorded in dripstone.

■ UPSC TRAPS

- ✗ The Meghalayan Age is a separate geological Epoch.
- ✓ It is an AGE/STAGE within the Holocene Epoch, not an epoch itself.

■ MAINS ANGLE

Show how karst dripstone serves as a palaeoclimate proxy that anchors the geological time scale.

■ STUDY LINK

Geography → Karst → Palaeoclimate / GSSP (applied CA)